BIRZEIT UNIVERSITY

Computer Science Department

COMP233-Discrete Mathematics / Spring 2026

Section #4 — Quiz#2 — Chapters 7,8.1

Time: 40 Minutes — **Total Marks:** 100 — **weight:** (10)

Student Name: ____________________ **Student ID:** ____________________

Answer Sheet

Answer 1

1. **Answer: b) f is one-to-one but not onto.**

   Let $f : \mathbb{Z} \to \mathbb{Z}$ be defined by $f(n) = 2n + 1$. If $f(a) = f(b)$, then $2a + 1 = 2b + 1$, so $2a = 2b$ and hence $a = b$. Therefore, f is one-to-one.

   However, f is not onto because $f(n) = 2n + 1$ is always odd. For example, there is no $n \in \mathbb{Z}$ such that $2n + 1 = 0$. Thus,

   $$\boxed{f \text{ is one-to-one but not onto.}}$$

2. **Answer: a) $2^3 = 8$.**

   Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$. A function from A to B assigns to each element of A exactly one element of B. Since $|A| = 3$ and $|B| = 2$, the number of functions is

   $$|B|^{|A|} = 2^3 = 8.$$

   The eight functions are:

	1	2	3
f_1	a	a	a
f_2	a	a	b
f_3	a	b	a
f_4	a	b	b
f_5	b	a	a
f_6	b	a	b
f_7	b	b	a
f_8	b	b	b

   Equivalently,

   $$\begin{aligned} f_1 &= \{(1,a),(2,a),(3,a)\}, & f_2 &= \{(1,a),(2,a),(3,b)\},\\ f_3 &= \{(1,a),(2,b),(3,a)\}, & f_4 &= \{(1,a),(2,b),(3,b)\},\\ f_5 &= \{(1,b),(2,a),(3,a)\}, & f_6 &= \{(1,b),(2,a),(3,b)\},\\ f_7 &= \{(1,b),(2,b),(3,a)\}, & f_8 &= \{(1,b),(2,b),(3,b)\}. \end{aligned}$$

   Therefore,

   $$\boxed{2^3 = 8.}$$

3. **Answer: b) Range$(f) = B$.**

   A function $f : A \to B$ is onto if every element of B is the image of at least one element of A; that is,

   $$\forall b \in B,\ \exists a \in A \text{ such that } f(a) = b.$$

   This is equivalent to saying Range$(f) = B$. Therefore,

   $$\boxed{\text{Range}(f) = B.}$$

4. **Answer: a) one-to-one but not onto.**

   Let $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ be defined by $f(m, n) = (m + n, m - n)$.

   To prove that f is one-to-one, suppose $f(m_1, n_1) = f(m_2, n_2)$. Then

   $$m_1 + n_1 = m_2 + n_2, \qquad m_1 - n_1 = m_2 - n_2.$$

   Adding gives $2m_1 = 2m_2$, so $m_1 = m_2$. Substituting back gives $n_1 = n_2$. Thus $(m_1, n_1) = (m_2, n_2)$, so f is one-to-one.

   To test whether f is onto, let $(a, b) \in \mathbb{Z} \times \mathbb{Z}$ and solve $(m + n, m - n) = (a, b)$. Then

   $$m + n = a, \qquad m - n = b,$$

   so

   $$m = \frac{a + b}{2}, \qquad n = \frac{a - b}{2}.$$

   These are not always integers. For example, if $(a, b) = (1, 0)$, then $m = \frac{1}{2}$ and $n = \frac{1}{2}$, which are not in $\mathbb{Z}$. Hence, f is not onto. Therefore,

   $$\boxed{f \text{ is one-to-one but not onto.}}$$

5. **Answer: d) neither one-to-one nor onto.**

   Let $F : \mathcal{P}(\{1, 2, 3\}) \to \mathbb{Z}^{\text{nonneg}}$ be defined by $F(S) = |S|$. The outputs of F can only be $0, 1, 2, 3$.

   The function is not one-to-one because

   $$F(\{1\}) = 1 = F(\{2\}), \qquad \{1\} \neq \{2\}.$$

   Also, F is not onto because the codomain is $\mathbb{Z}^{\text{nonneg}} = \{0, 1, 2, 3, 4, 5, \ldots\}$, but no subset of $\{1, 2, 3\}$ has cardinality 4 or larger. Therefore,

   $$\boxed{F \text{ is neither one-to-one nor onto.}}$$

Answer 2

(a)

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 4x - 1$.

Definitions. A function $f : A \to B$ is one-to-one if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. It is onto if $\forall y \in B$, $\exists x \in A$ such that $f(x) = y$. A one-to-one correspondence is a function that is both one-to-one and onto.

To prove one-to-one, assume $f(x_1) = f(x_2)$. Then $4x_1 - 1 = 4x_2 - 1$, so $4x_1 = 4x_2$ and hence $x_1 = x_2$. Therefore, f is one-to-one.

To prove onto, let $y \in \mathbb{R}$. We need $x \in \mathbb{R}$ such that $f(x) = y$. Solving $4x - 1 = y$ gives $x = \frac{y+1}{4}$, which is a real number. Therefore, f is onto. Hence,

$$\boxed{f \text{ is a one-to-one correspondence.}}$$

(b)

General definition of inverse function. Let $f : A \to B$ be a function. The inverse function $f^{-1} : B \to A$ exists if and only if f is a one-to-one correspondence. If $y = f(x)$, then $x = f^{-1}(y)$. Equivalently,

$$f^{-1}(f(x)) = x \qquad \text{and} \qquad f(f^{-1}(y)) = y.$$

Assumption for this example. From part (a), $f(x) = 4x - 1$, and since f is a one-to-one correspondence, the inverse function exists.

Let $y = f(x)$. Then $y = 4x - 1$. Interchange x and y to get $x = 4y - 1$. Solving for y gives $x + 1 = 4y$, so

$$y = \frac{x + 1}{4}.$$

Therefore,

$$\boxed{f^{-1}(x) = \frac{x+1}{4}}.$$

Verification:

$$f(f^{-1}(x)) = f\left(\frac{x+1}{4}\right) = 4\left(\frac{x+1}{4}\right) - 1 = x + 1 - 1 = x.$$

Thus,

$$\boxed{f(f^{-1}(x)) = x.}$$

Also,

$$f^{-1}(f(x)) = f^{-1}(4x-1) = \frac{(4x-1)+1}{4} = \frac{4x}{4} = x.$$

Therefore,

$$\boxed{f^{-1}(f(x)) = x.}$$

(c)

Let $g : \mathbb{Z} \to \mathbb{Z}$ be defined by $g(n) = n^2$. A function $g : A \to B$ is one-to-one if $g(a_1) = g(a_2) \Rightarrow a_1 = a_2$. To disprove one-to-one, it is enough to give two different inputs with the same output.

Take $a_1 = 1$ and $a_2 = -1$. Then

$$g(1) = 1^2 = 1, \qquad g(-1) = (-1)^2 = 1.$$

Thus, $g(1) = g(-1)$ while $1 \neq -1$. Therefore,

$$\boxed{g \text{ is not one-to-one.}}$$

(d)

Consider the proposed function $w : \mathbb{Q} \to \mathbb{Q}$ defined by $w\left(\frac{m}{n}\right) = m + n$, where $m, n \in \mathbb{Z}$ and $n \neq 0$.

A function is well-defined if each input has exactly one output. For functions defined on rational numbers, the value must not depend on the particular fraction representation of the same rational number.

A function $f : A \to B$ is a mapping from the domain A to the codomain B in which every element of the domain has *exactly one* outgoing arrow to an element of the codomain. Several elements of the domain may point to the same element of the codomain, and some elements of the codomain may receive no arrows. However, no element of the domain may have two different images or no image at all. Therefore, when the domain is $\mathbb{Q}$, each rational number is a single domain element, regardless of its representation. Hence, equivalent fractions such as $\frac{1}{2} = \frac{2}{4}$ must map to the same output; otherwise, the assignment is not a well-defined function.

Since $\frac{1}{2} = \frac{2}{4}$, we compute

$$w\left(\frac{1}{2}\right) = 1 + 2 = 3, \qquad w\left(\frac{2}{4}\right) = 2 + 4 = 6.$$

But $3 \neq 6$. Thus, the same rational number gives two different outputs. Therefore,

$$\boxed{w \text{ is not well-defined.}}$$

Answer 3

(a)

Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$. Define the relation R from A to B by $xRy \iff x \mid y$. Thus, $(x, y) \in R$ if and only if x divides y.

For $x = 2$, we have $2 \mid 2$, $2 \mid 6$, and $2 \mid 8$, giving $(2, 2), (2, 6), (2, 8)$. For $x = 3$, only $3 \mid 6$, giving $(3, 6)$. For $x = 4$, only $4 \mid 8$, giving $(4, 8)$. Therefore,

$$\boxed{R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8)\}.}$$

(b)

If R is a relation from A to B, then the inverse relation R^{-1} from B to A is defined by

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}.$$

Thus, we reverse each ordered pair in R:

$$\boxed{R^{-1} = \{(2, 2), (6, 2), (8, 2), (6, 3), (8, 4)\}.}$$

(c)

The arrow diagram of R^{-1} is from B to A. Since $R^{-1} = \{(2, 2), (6, 2), (8, 2), (6, 3), (8, 4)\}$, the arrows are

$$2 \to 2, \qquad 6 \to 2, \qquad 6 \to 3, \qquad 8 \to 2, \qquad 8 \to 4.$$

Equivalently,

$$\boxed{2 \mapsto 2, \qquad 6 \mapsto 2, 3, \qquad 8 \mapsto 2, 4.}$$